

PROJECT PROPOSAL (PROPOSED SOLUTION)

ComicCraft – AI Comic Story Creator using Gemini Models

Date	Not specified in the ComicCraft project document
Team ID	Not specified in the ComicCraft project document
Project Title	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposal is to develop ComicCraft, a web-based AI application that transforms user-provided creative ideas into personalized comic stories and illustrations. The system combines Google Gemini models for structured story generation, narration and dialogue with Stable Diffusion for comic-style image generation. The complete comic can be previewed through a web interface and exported as a downloadable PDF.

Project Overview

Objective	To simplify comic creation by automatically generating personalized, panel-by-panel stories, character dialogues, scene descriptions and comic-style illustrations from user prompts.
Scope	The project covers user input collection, AI-based outline generation, story and dialogue generation, image generation, comic layout creation, web preview and PDF export. The current version focuses on single-comic generation without user account management.

Problem Statement

Description	Creating a complete comic traditionally requires separate skills for story development, dialogue writing, illustration and page layout. The project addresses this fragmented and time-consuming process by providing an automated AI-assisted workflow from a simple creative prompt to a complete comic.
Impact	The proposed system makes comic creation more accessible to non-artists, storytellers and hobbyists by reducing the manual effort required to produce a coherent visual story. It also allows users to regenerate content with different tones and art styles.

Proposed Solution

Approach	Employ a modular FastAPI-based web application that receives the user's story prompt and preferences, invokes Gemini Flash to create a structured 5-panel outline, uses Gemini Pro to expand the outline into narration and dialogue, and uses Stable Diffusion to create an illustration for each panel. The generated assets are then organized and exported as a multi-page PDF.
Key Features	• Personalized comic generation from user prompts. • 5-panel structured outline generation using Gemini Flash. • Detailed narration and character dialogue using Gemini Pro. • Comic-style illustration generation using Stable Diffusion. • Panel-by-panel comic preview through Jinja2 templates. • PDF export using FPDF. • Support for different story tones and art styles. • JSON API endpoint for programmatic comic generation.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing resources required for AI-based image generation and application execution.	Python-compatible development system; GPU support may be used for Stable Diffusion. Exact hardware specification is not stated in the project document.

Resource Type	Description	Specification / Allocation
Software	Application framework, AI libraries, image generation and PDF generation tools.	Python, FastAPI, Uvicorn, Jinja2, python-multipart, google-generativeai, Diffusers, Transformers, FPDF, Pillow, Accelerate, PyTorch
Development Environment	Environment for dependency isolation, application execution and API testing.	Python virtual environment; pip; local Uvicorn server; FastAPI /docs
AI Services	Text and image generation services used in the comic creation pipeline.	Google Gemini Flash, Google Gemini Pro, Hugging Face Diffusers / Stable Diffusion
Storage	Stores generated panel images and exported comic PDFs.	static/panels and static/exports directories

Project Generation Workflow

Step	Stage	Implementation
1	User Input	Story prompt + character name + setting + tone + art style
2	Outline	Gemini Flash generates a structured 5-panel outline
3	Story	Gemini Pro generates narration and character dialogue
4	Images	Stable Diffusion generates comic-style illustration for each panel
5	Layout	build_comic_layout() matches images with panel story content
6	Export	save_pdf() uses FPDF to create the downloadable multi-page PDF
7	Preview	Jinja2 templates display the generated comic and export confirmation

Note: This document is prepared by mapping the supplied ComicCraft project information to the structure of the supplied Proposed Solution template. Fields not supported by the ComicCraft source document are explicitly marked as not specified rather than inferred.